Team [TODO: Team Number]

Problem [TODO: A or B]

[TODO: Paper Title]

Summary

[TODO: In at most 300 words, summarize the problem, approach, essential assumptions, numerical answers with units, and conclusions. Replace this entire paragraph.]

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1 Introduction

1.1 Problem Restatement

[TODO: Restate the selected problem in your own words. Define the physical setting, the questions to answer, and the scope and boundaries of the analysis.]

1.2 Background and Approach

[TODO: Briefly review relevant literature with in-text citations. Explain the physical mechanism and give a roadmap of the approach.]

2 Materials and Methods

2.1 Assumptions and Approximations

[TODO: State all assumptions and approximations; explain why each is appropriate using evidence, scale estimates, or comparison with the relevant physical regime.]

A1. **[TODO: Assumption, justification, and applicability.]**

A2. **[TODO: Assumption, justification, and applicability.]**

2.2 Physical Model

[TODO: Define the system, variables, units, governing laws, initial and boundary conditions. Justify the model and cite external inputs.]

2.3 Solution and Computational Method

[TODO: Explain derivation and numerical methods in the main text, including parameter values, sources, algorithms, convergence criteria, and any software citations as applicable.]

3 Results

3.1 Main Results

[TODO: Present answers to each question, with units and appropriate significant figures. Explain figures and tables in the text.]

3.2 Validation and Checks

[TODO: Check units and limiting cases; compare with an independent estimate, reference, experiment, or numerical convergence test where appropriate.]

4 Discussion

4.1 Sensitivity and Uncertainty

[TODO: Describe how uncertainties and changes in important assumptions or parameters affect the results.]

4.2 Strengths, Limitations, and Applicability

[TODO: Discuss strengths and weaknesses, neglected effects, and the conditions under which the model may fail.]

5 Conclusion

[TODO: Answer the original questions directly, summarize the principal numerical results and their limitations, and give the main takeaway.]

References

[1] **[TODO: Replace with a real reference and cite it in the text. Add more \bibitem entries as needed.]**

A Supplementary Material

[TODO: Optional: provide supporting derivations, extra data, or code. Explain essential methods, algorithms, and results in the main text.]